

K712 — PRE-REGISTERED PMNS audit rubric. Written BEFORE Grace’s Majorana render. Key reframe: PMNS angles are ALREADY BANKED (1/45, 3/10, 4/7); the new work is MECHANISM-UNIFICATION — does the rank-1-Dirac + Majorana-Takagi mechanism reproduce them? — not a from-scratch derivation. Plus the CKM lessons applied.

Keeper | 2026-07-16 | Arming the PMNS audit the way K705 armed CKM. Criteria fixed before the render so the verdict is mechanical. Incorporates: the banked-forms context, the $m_{\nu 1}=0$ large-angle mechanism, and every discipline lesson from the CKM close-out (K711).

THE REFRAME — PMNS is not from scratch (verified, pmns_check.py)

The corpus already banks the PMNS angle FORMS (D-tier), and they are excellent-to-structural: $-\sin^2\theta_{13} = 1/(N_c^2 \cdot n_C) = 1/45 = 0.0222$ vs obs $0.0222 \rightarrow 0.1\%$ ($\theta_{13}=8.6^\circ$). Excellent. $-\sin^2\theta_{23} = ((n_C-1)/(n_C+2)) \cdot \cos^2\theta_{13} = (4/7)\cos^2 = 0.559$ vs obs $0.572 \rightarrow 2.3\%$ ($\theta_{23}=48.4^\circ$). Structural-good. $-\sin^2\theta_{12} = (N_c/2n_C) \cdot \cos^2\theta_{13} = (3/10)\cos^2 = 0.293$ vs obs $0.307 \rightarrow 4.5\%$ ($\theta_{12}=32.8^\circ$). Structural. So the DELIVERABLE is mechanism-unification, not a new number: does the same mechanism that closed CKM (one Higgs $\rightarrow$ rank-1 Dirac for charged leptons; Majorana/Takagi for neutrinos) FORWARD-REPRODUCE 1/45, 3/10, 4/7? If yes $\rightarrow$ PMNS upgrades from “banked forms” to “banked forms WITH a derived mechanism” (a real strengthening). If it gives different values $\rightarrow$ a tension between mechanism and banked forms to resolve (don’t paper over).

THE LARGE-ANGLE MECHANISM — verify this is what Grace’s render uses (target-innocent)

- Charged leptons: Dirac, rank-1 Yukawa (one Higgs) $\rightarrow U_\ell$ near-diagonal, small angles (like the quarks). $\square$ established.
- Neutrinos: Majorana $\rightarrow$ complex-SYMMETRIC mass matrix $\rightarrow$ Takagi ($U \square MU$), NOT Hermitian (K706 showed Hermitian gives ≈ 0 ; Takagi swung θ_{23} $0.4^\circ \rightarrow 53^\circ$). And $m_{\nu 1} = 0$ (banked, $\mathbb{Z}_3$ -protected) leaves a FLAT direction $\rightarrow$ the mixing is UNLOCKED from the mass ratios $\rightarrow$ naturally LARGE. This is the PMNS-large mechanism, and it is target-innocent (rests on $m_{\nu 1}=0$ + Majorana, both banked before mixing).
- PMNS = $U_\ell^\dagger U_\nu \approx U_\nu$ (charged-lepton rotation small) $\rightarrow$ inherits the large neutrino angles.

PRE-REGISTERED PASS/FAIL (bands fixed NOW)

- PASS (mechanism-unification): the rank-1-Dirac + Majorana-Takagi render forward-reproduces the three angles within $\sim 15\%$ of the banked forms ($\theta_{13} \approx 8.6^\circ$, $\theta_{12} \approx 33^\circ$, $\theta_{23} \approx 45-49^\circ$), with θ_{13} clearly nonzero (rules out exact tribimaximal — see trap), from the $m_{\nu 1}=0$ /Majorana structure, NOT fit.
- CONDITIONAL: two of three reproduce, the third misses but fingers a named piece (the neutrino locus F413 / the $d=5 \rightarrow d=4$ projection radius).
- FAIL: the render only gets large angles by an imposed ansatz (TBM, anarchy) rather than forward from the Majorana structure; or $\theta_{13}=0$ (tribimaximal — excluded by data).

TARGET-INNOCENCE TRAPS SPECIFIC TO PMNS (Cal fires here)

1. The tribimaximal trap. Large PMNS angles are EASY to fit — TBM ($\theta_{12}=35.3^\circ$, $\theta_{23}=45^\circ$, $\theta_{13}=0$) is the classic ansatz, and exact TBM is now EXCLUDED by $\theta_{13}=8.6^\circ \neq 0$. So BST must produce $\theta_{13} \neq 0$ FORWARD — which 1/45 does (that’s the strong one). Any route that imposes maximal/TBM mixing by hand and adds θ_{13} as a correction is a fit, not a derivation.
2. The anarchy trap. “Large neutrino mixing is generic/anarchic” explains largeness but predicts nothing. BST’s angles are SPECIFIC (1/45, 3/10, 4/7) — the mechanism must give those specific values, not just “large.”

3. The sub-percent trap (CKM lesson, K711). $\theta_{13}=1/45$ lands 0.1% — genuinely tight — but θ_{12} , θ_{23} are structural (2–5%). Do NOT chase sub-percent on θ_{12}/θ_{23} by tuning radii; the genus-power sensitivity makes near-fits meaningless (the K711 fish). Structural is the honest tier for the large angles.

δ_{PMNS} — the one genuinely OPEN number (pre-register as a PREDICTION)

δ_{PMNS} is unconstrained by current data (DUNE/HyperK will measure it). The rank-1/FK-“1–r_i r_j· ω ”-twist CP mechanism (Elie, the lepton analog of the CKM J) should give it FORWARD. Pre-register: whatever δ_{PMNS} the mechanism produces is a falsifiable PREDICTION (DUNE-testable), banked as a prediction only if it comes from the same FK-twist that gave the CKM J — not a separate input. Do not fit it to any current central value (there isn’t a pinned one).

MISS PROTOCOL

If the mechanism gives angles $\neq$ the banked 1/45, 3/10, 4/7: this is a mechanism-vs-banked-form tension — do NOT silently adopt whichever matches data better. Report both, and resolve by which is target-innocent (the banked forms are D-tier; the mechanism is newer). The neutrino locus (F413, d=5→d=4 projection) is the likely reconciliation point — the analog of the CKM projection radius.

WHAT WOULD BANK

- PMNS mechanism-unification: if the rank-1/Majorana-Takagi mechanism reproduces 1/45, 3/10, 4/7 forward → the three angles upgrade to “banked form + derived mechanism,” and PMNS joins CKM as mechanism-closed.
- δ_{PMNS} : a forward prediction from the FK-twist (falsifiable at DUNE), banked as prediction.
- NOT banked: any angle obtained by TBM/anarchy ansatz, or sub-percent tuning on θ_{12}/θ_{23} .

— Keeper K712, 2026-07-16. PMNS PRE-REGISTERED: angles ALREADY banked (1/45 at 0.1%, 3/10, 4/7); deliverable = mechanism-unification (does rank-1-Dirac + Majorana-Takagi reproduce them?). Large-angle mechanism = $m_{\nu 1}=0$ unlocks mixing from mass ratios (target-innocent). Traps: tribimaximal (θ_{13} must be forward-nonzero — 1/45 delivers), anarchy, sub-percent tuning (K711 lesson). δ_{PMNS} = the one open number, pre-registered as a DUNE-testable prediction from the FK-twist. See [[Keeper_K711...]] (CKM close), [[Keeper_K706...]] (Takagi lead).